

| Question | Answer                                                                                                                                                                         | Marks     |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 7(a)     | <ul style="list-style-type: none"> <li>• (electron) neutrino</li> <li>• (electron) antineutrino</li> <li>• electron</li> <li>• positron</li> </ul> <p>Any two, 1 mark each</p> | <b>B2</b> |
| 7(b)     | only meson <u>and</u> neutron are underlined                                                                                                                                   | <b>B1</b> |
| 7(c)     | <p>(quark composition is) up, up, down / uud</p> <p>(charge =) <math>\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = (+1)e = (+) 1.6 \times 10^{-19} \text{ (C)}</math></p>       | <b>B1</b> |
|          | <p><i>or</i></p> <p>(charge =) <math>1.07 \times 10^{-19} + 1.07 \times 10^{-19} - 5.33 \times 10^{-20} = (+) 1.6 \times 10^{-19} \text{ (C)}</math></p>                       | <b>A1</b> |